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10 / 10 submitted

Q1. Calculate the Total, Average, and Percentage

ANSWER

```
BEGIN
step 1: Begin the program
step 2 : input the marks of Subject 1,Subject 2, and Subject 3
step 3 : Calculate the Total by adding the three Subject marks.
step 4: calculate the Average by dividing the Total by 3
step 5: Calculate the Percentage using the formula
        percentage=(total/300) * 100
step 6: Display the total
step 7: Display the average
step 8: Display the Percentage
step 9: End the program
END
```

Q2. Celsius temperature into Fahrenheit

ANSWER

```
BEGIN
Step 1: Begin the program.
Step 2: Input the temperature in celsius
Step 3: Convert the temperature to Fahrenheit using the formula :
        fahrenheit = (celsius * 9/5) + 32
step 4 : Display the temperature in fahrenheit
Step 5: End the program
END
```

Q3. extract the last three characters of a given string

ANSWER

```
BEGIN
Step 1: Begin the program
Step 2: Read the input string from the user
Step 3: store the string
Step 4: Find the length of the string
Step 5: Extract the last three characters from the string
Step 6: Store the extracted Characters
Step 7: Display the last three Characters
Step 8: End the program
END
```

Q4. Area and Perimeter of Rectangle

CODE

```
BEGIN
INPUT Length,Breadth
Area ←Length + Breadth
Perimeter ← 2 *(Length * Breadth)
Print (Area)
Print (Perimeter)
END
```

OUTPUT

Q5. convert days into years, months, and days

CODE

```
BEGIN
INPUT Days
Years ← Days DIV 365
RemainingDays1 ← Days MOD 365
Months ← RemainingDays1 div 30
Remainingdays2 ← RemainingDays1 MOD 30
Print years
Print Months
Print Remainingdays2
END
```

INPUT

400

OUTPUT

Q6. identify whether a character is an alphabet, digit, or special character.

CODE

```
BEGIN
Ch=input()
if(Ch ≥ 'A' and Ch ≤ 'z') or (ch ≥ 'a' and Ch ≤ 'z'):
    print("Alphabet")
else if Ch ≥ '0' and Ch ≤ '9':
    print("Digit")
else
    print("special character")
END
```

INPUT

A

OUTPUT

Q7. Simple Interest

ANSWER

 $SI = (\text{principal} * \text{Rate} * \text{Time}) / 100$ **Q8. swap two numbers**

ANSWER

Temp=A , A=B , B=temp

Q9. Check Whether Two Strings are Equal

CODE

```
BEGIN
Str1 = "python"
Str2 = "Python"
if str1==str2:
    print("equal")
else:
    print("Not equal")
END
```

OUTPUT

Q10. Convert Days into Months and Remaining Days

CODE

```
BEGIN
days=int(input())
months = days // 30
remainingdays = days % 30
print(months)
print(remainingdays)
END
```

INPUT

75

OUTPUT

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Q1. Harshad number

ANSWER

```
n = int(input())

s = sum(int(i) for i in str(n))

if n % s == 0:
    print("harshad number")
else:
    print("not a harshad number")

print("harshad numbers from 1 to 500:")
for n in range(1,501):
    if n % sum(int(i) for i in str(n)) == 0:
        print(n,end=" ")
```

Q2. GCD

CODE

```
a = int(input("Enter the first:\n"))
b = int(input("Enter the second:\n"))

while b != 0:
    temp = b
    b = a % b
    a = temp

print("Gcd is",a)
```

INPUT

24
36

OUTPUT

Enter the first:
Enter the second:
Gcd is 12**Q3. sum to a target value**

ANSWER

```
def find_pairs(lst,target):
    seen = set()
    pairs = []
    for num in lst:
        complement = target - num

        if complement in seen:
            pairs.append((complement,num))

        seen.add(num)

    return pairs

data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = find_pairs(data,t)
print("pairs:",result)
```

Q4. Bitwise Operators

ANSWER

```
a = 13
b = 21
c = 26
result = 31
0b11111
0x1f
7 62 -32
```

Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low = 0
    high = len(arr)-1

    while low ≤ high:
        mid = (low + high)// 2

        if arr[mid] == target:
            return mid

        if arr[low] ≤ arr[mid]:
            if arr[low] ≤ target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1

        else:
            if arr[mid] < target ≤ arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1
data = [4,5,6,7,0,1,2]
target = int(input("Target: "))

result = rotated_binary_search(data,target)
print("Found at:",result)
```

OUTPUT

Target: Found at: 4